

Basic Principles of Medicine II
Module: Nervous System and Behavioral
Sciences
Lecture No: NB 07

Excitable Cells

Dr Duncan Kirkby

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Recommended Reading

[Electrophysiology of Neurons | Essential Neuroscience, 4e | Medical Education | Health Library \(oclc.org\)](#)

Abbreviations

AP – Action Potential

CNS – Central Nervous System

IIZ - impulse initiation zone

Na_v – voltage-gated sodium channel

λ – length constant (lambda)

E_{ion} – Equilibrium potential / Nernst potential / Diffusion potential

PNS – peripheral nervous system

V_m – membrane potential (measured)

Ions:

H^+ – hydrogen

K^+ – potassium

Na^+ – sodium

Cl^- – chloride

Learning Objectives

- SOM.MK.I.BPM2.1.NB.1.NSCI.0026 Review the relation between the stimulus delivered to an excitable cell and the associated graded electrical response.
- SOM.MK.I.BPM2.1.NB.1.NSCI.0027 Explain how excitable cells convert stimuli into responses consisting of changes of conductance and membrane potential.
- SOM.MK.I.BPM2.1.NB.1.NSCI.0028 Explain the concept of electrical signals spreading from their sites of origin with decrement, governed by a length constant (λ).
- SOM.MK.I.BPM2.1.NB.1.NSCI.0029 Relate the electrical properties of cable-like structures, such as axons and dendrites, to their length constants.

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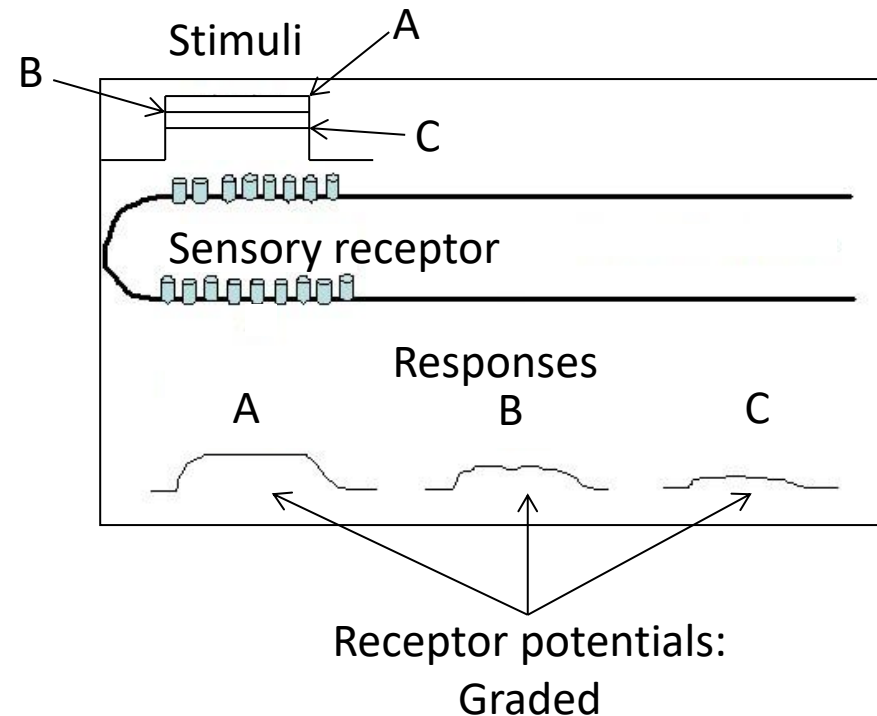
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Excitable Cells

- Excitable cells = cells that can express action potentials
 - Neurons (most)
 - Action potentials are critical for neural communication
 - Neurons commonly integrate inputs and fire action potentials to trigger release of neurotransmitter
 - Muscle
 - Skeletal: contraction driven by Na^+ -mediated action potentials
 - Cardiac: myocyte contraction, sinoatrial (SA) node generates action potentials to set pacemaker rhythm
 - Smooth: Ca^{2+} mediates depolarisation and contraction
 - Some endocrine cells
 - e.g., pancreatic β cells, pituitary

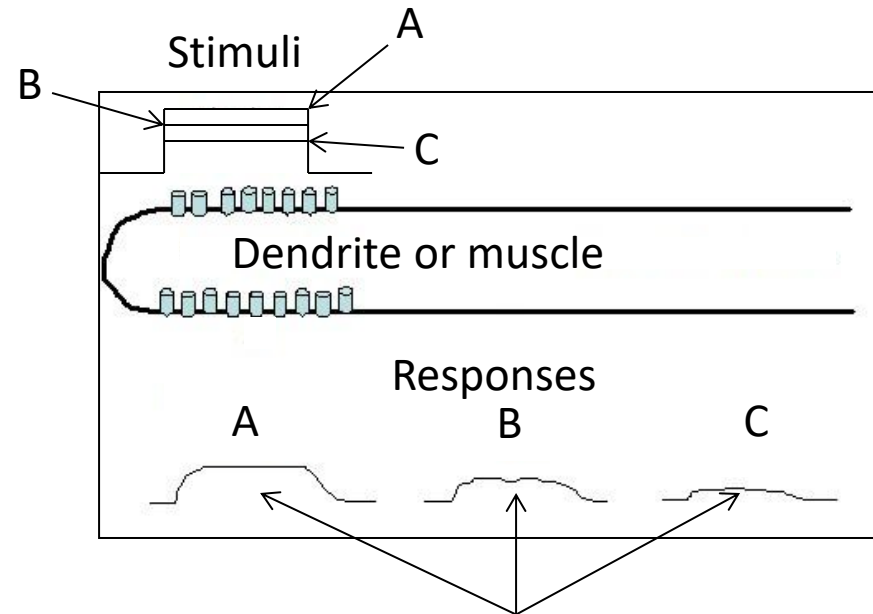
Stimulus-Response Characteristics

- Receptive endings of neurons respond electrically to specific stimuli
 - Ion channels open in response to pressure, heat, etc.
- Local receptor potentials are graded
 - V_m change varies with the magnitude of the stimulus
 - E.g., more pressure = bigger graded receptor potential



Signal Transduction

- Chemical receptors for specific stimuli are linked to ion channels
 - Direct link: Receptor and channel are part of the same complex, e.g., ACh receptor at neuro-muscular junction (fast)
 - Indirect link: Via 2nd messenger systems (slower)
 - Stimulation causes change in ion conductance and hence V_m
- Post-synaptic potentials (graded)
 - Arise in post-synaptic membranes receiving chemical inputs from pre-synaptic neurons



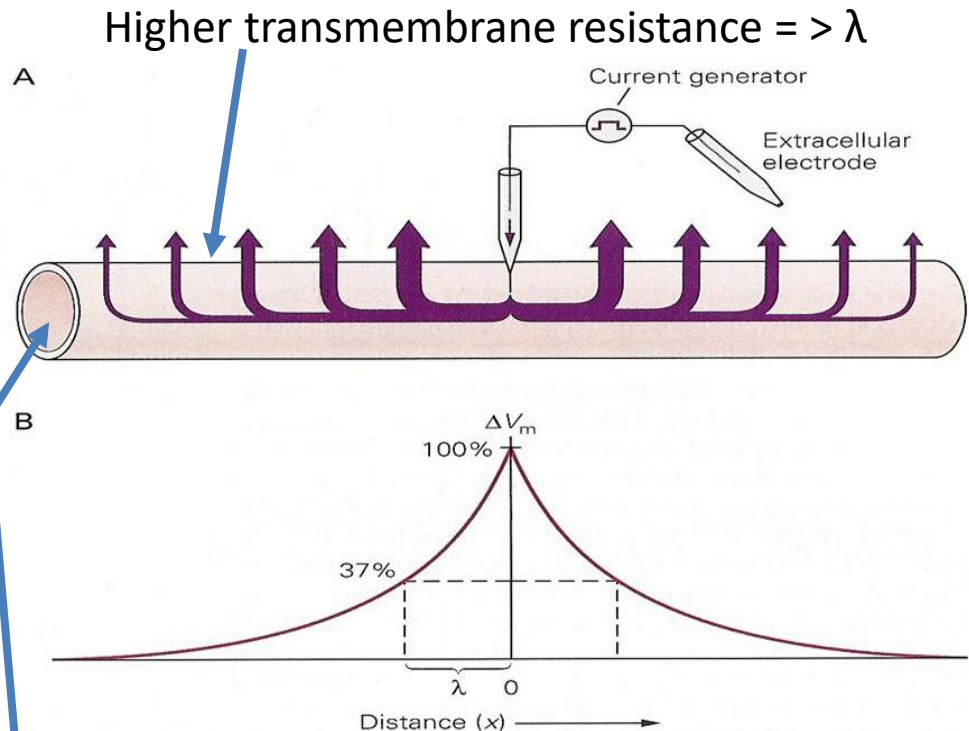
Neuronal postsynaptic potentials: Graded
or
Muscular endplate potentials: Graded

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Length constant

- Receptor potentials and postsynaptic potentials are graded decremental deviations from resting membrane potential (i.e., they vary in magnitude and diminish with increased distance from the site of initiation)
- Length constant λ (lambda)
 - Distance over which a graded potential decays to 37% (1/e) of its maximal amplitude
 - Greater λ means less decrement over distance
 - Depends on intracellular (r_i) and transmembrane resistance (r_m)
 - Typical λ values for neurons are 0.1-1.5 mm, i.e., depolarization spreads locally only over short distances



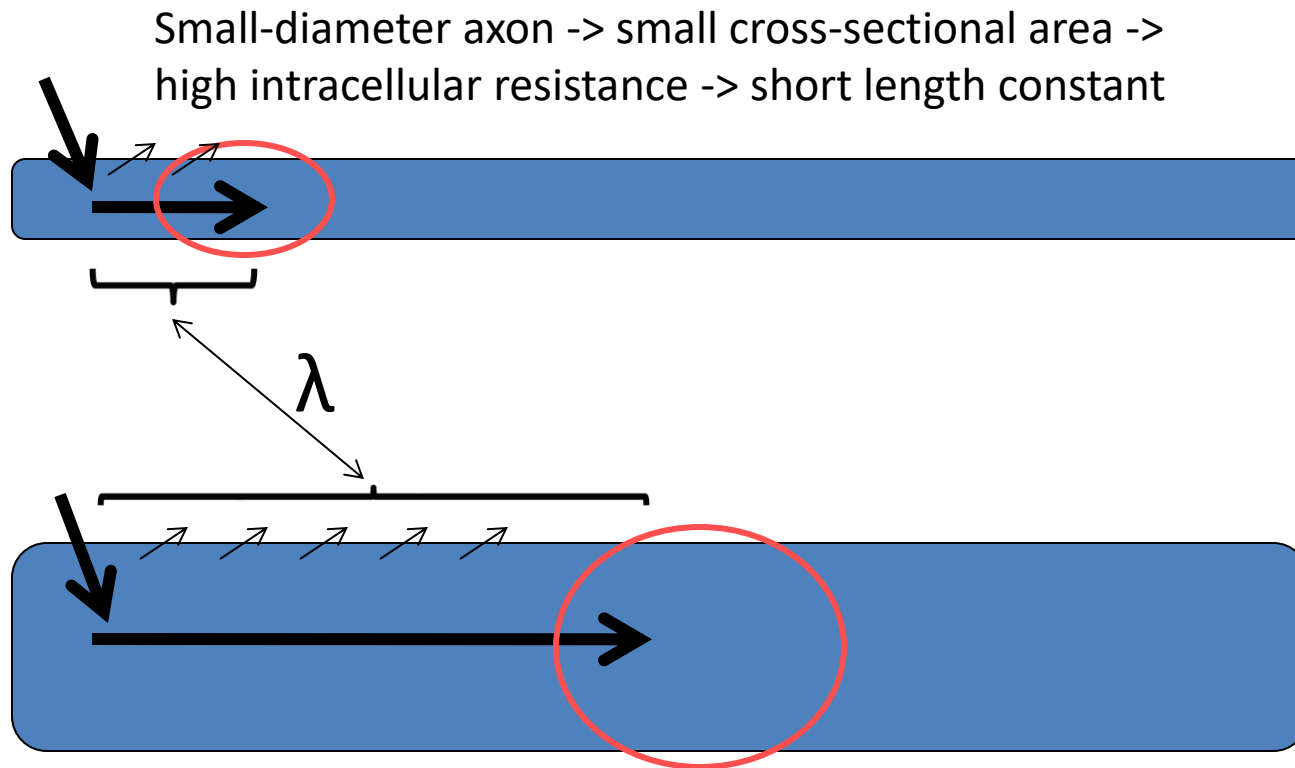
Lower intracellular resistance $\Rightarrow \lambda$

$$\lambda = \sqrt{\frac{r_m}{r_i}}$$

Length constant and cable properties

- Compared to small axons, larger diameter axons have
 - Internal cross-sectional area
 - Larger
 - Intra-axonal resistance to ion flux
 - Less
 - Length constant
 - Greater
 - Conduction speed
 - Greater
- Examples (diameter and conduction velocity shown)
 - Human small unmyelinated fibre: 0.2 – 1.5 μm , 0.5 - 2 m/s
 - Squid giant axon: 0.5 – 1 mm, 20 - 25 m/s

Conduction Velocity and Axonal Diameter



Conduction Velocity and Axonal Diameter

- Small-diameter axons have a small cross-sectional area, which increases intracellular electrical resistance (r_i)
 - The increased denominator necessitates a decreased length constant

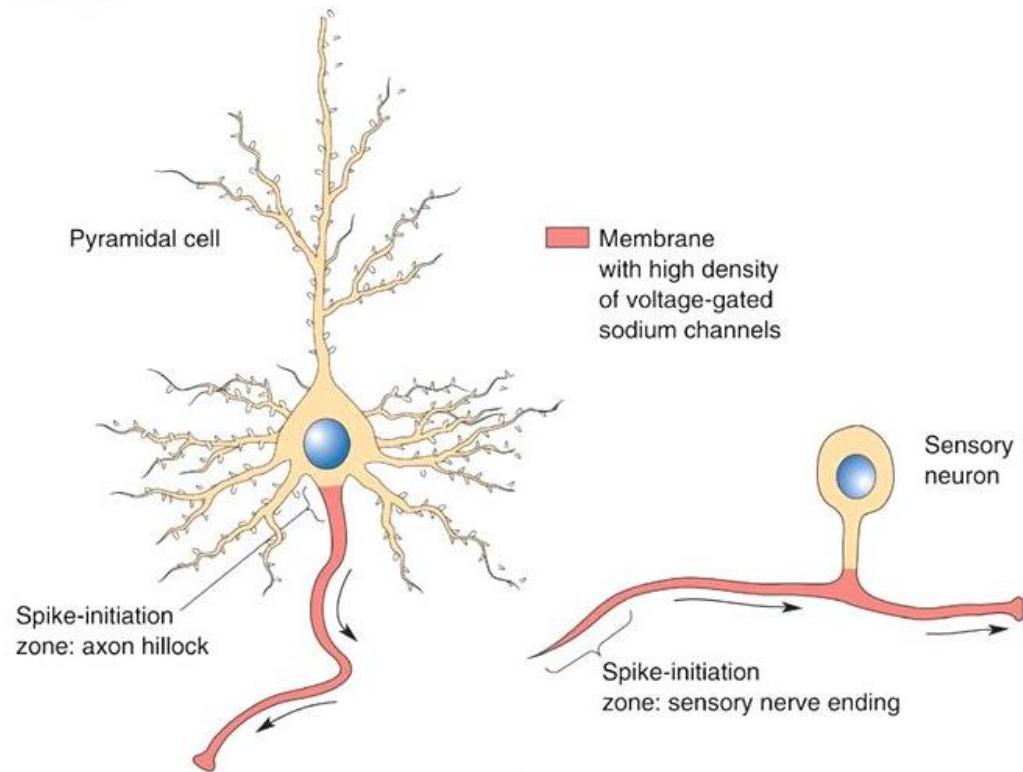
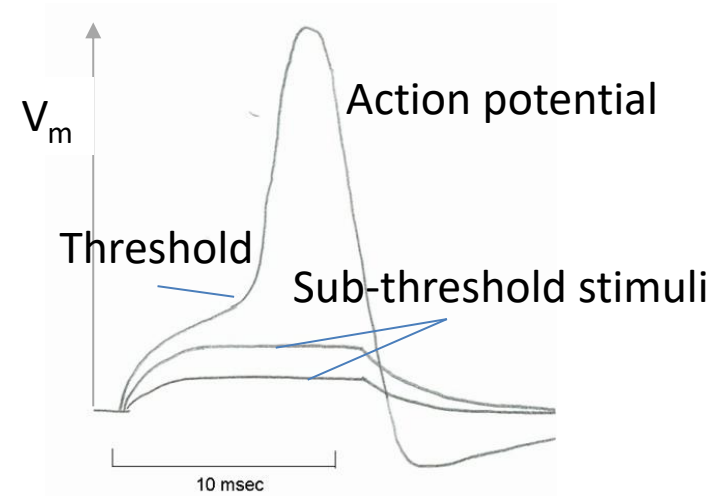
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Action potential initiation

- Input integration
 - Graded potentials generate action potentials if their sum exceeds the depolarization threshold at a special axonal region
 - Threshold = voltage that is sufficient to open voltage-gated channels
- Impulse Initiation Zone (IIZ)
 - Near axonal hillock (at start of axon), or near sensory nerve endings in sensory neurons
 - Contains the highest density of voltage-gated Na^+ channels
 - Most excitable part of neuron
 - Site of AP initiation



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Na_v structure-function relationship

- 4 domains, each having 6 transmembrane segments
- S4 is positively charged, confers voltage sensitivity
- S5 and S6 line the inner ion pore
- Rings of amino acids around pore confer ion selectivity
- Intracellular loops: Inactivation gate (D3-D4 short link) and phosphorylation sites (long intracellular links)

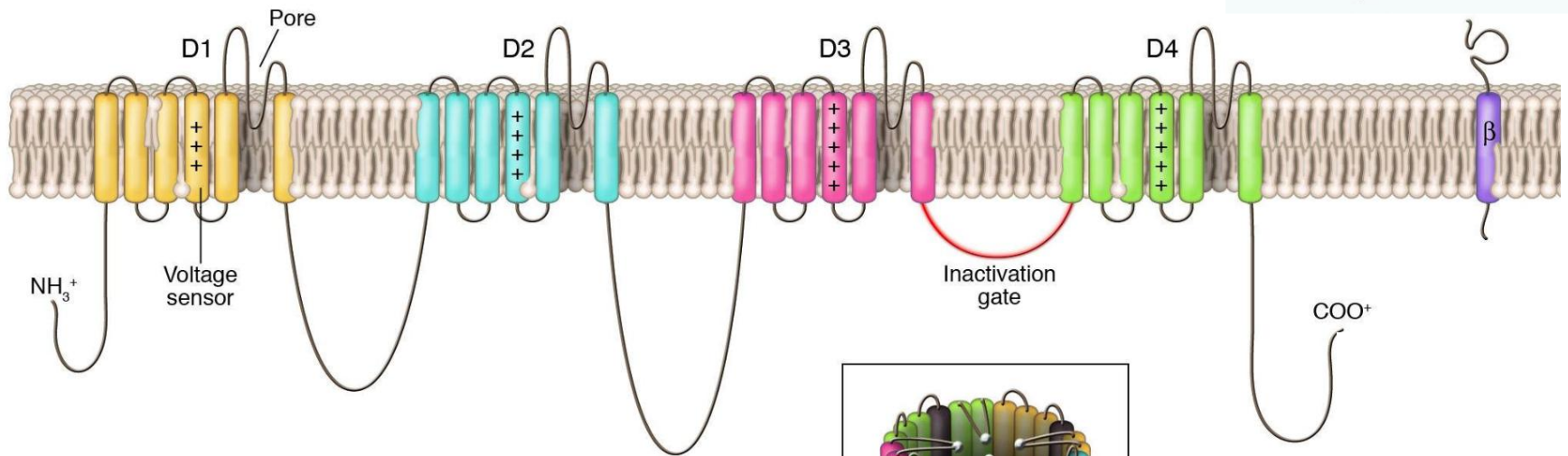
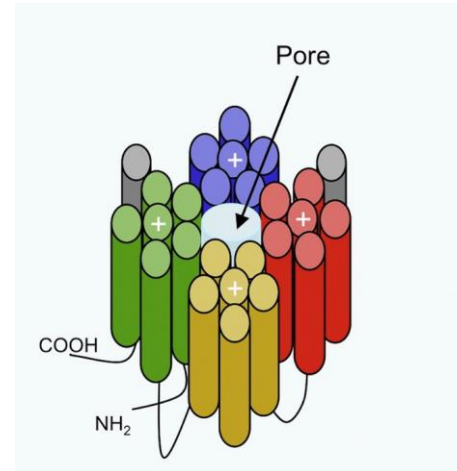


Image from Meisler, M. H. (2005). Sodium channel mutations in epilepsy and other neurological disorders. *Journal of Clinical Investigation*, 115(8), 2010–2017.

<https://doi.org/10.1172/JCI25466>

NSCI.0031

Na_v activation gate

The S4 segment confers voltage sensitivity: Positively charged amino acids serve as the activation (m) gate and are repelled outwards by increased internal positivity tied to depolarization.

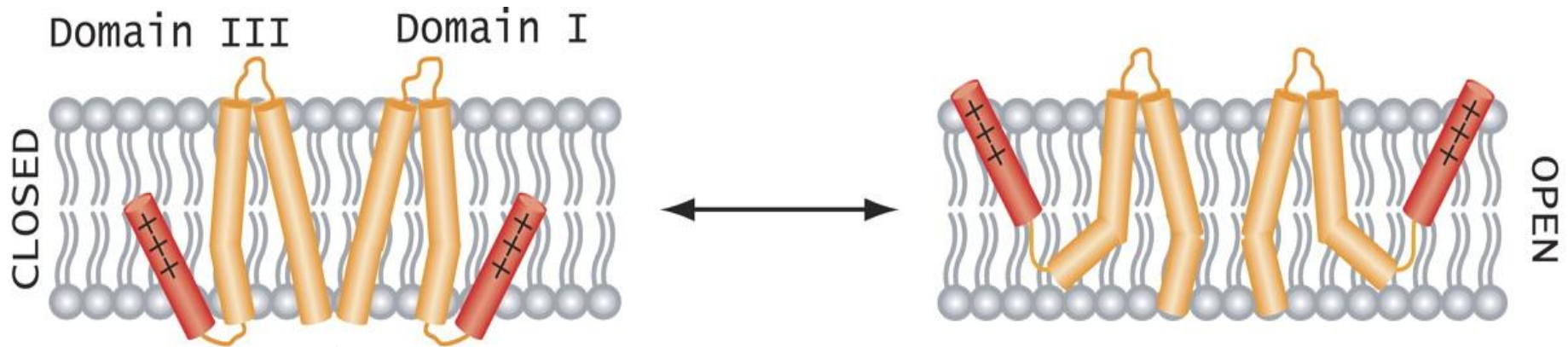
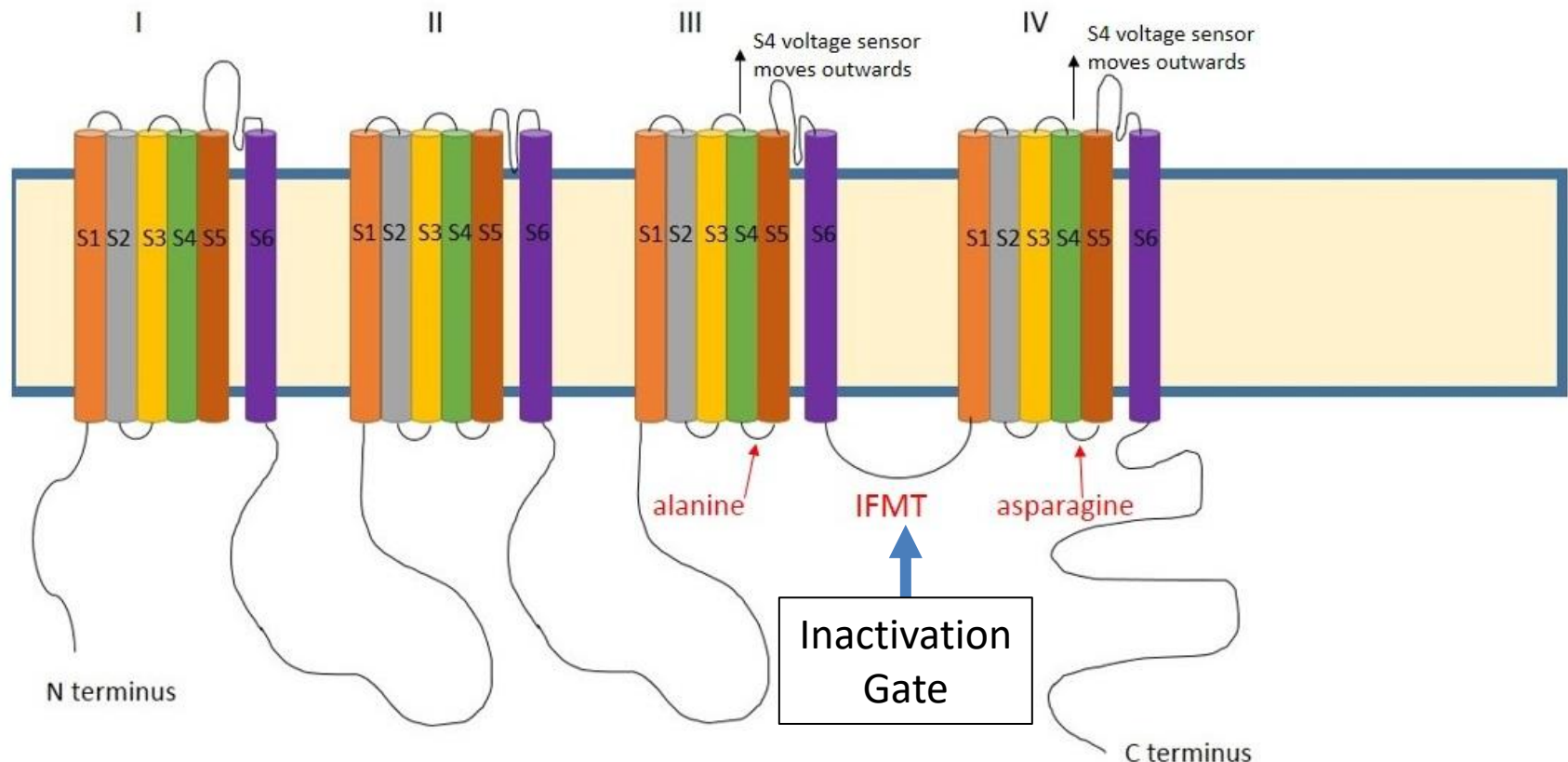


Image modified from Arcisio-Miranda, M., Muroi, Y., Chowdhury, S., & Chanda, B. (2010). Molecular mechanism of allosteric modification of voltage-dependent sodium channels by local anesthetics. *The Journal of General Physiology*, 136(5), 541–554. <https://doi.org/10.1085/jgp.201010438>

Na_v inactivation gate

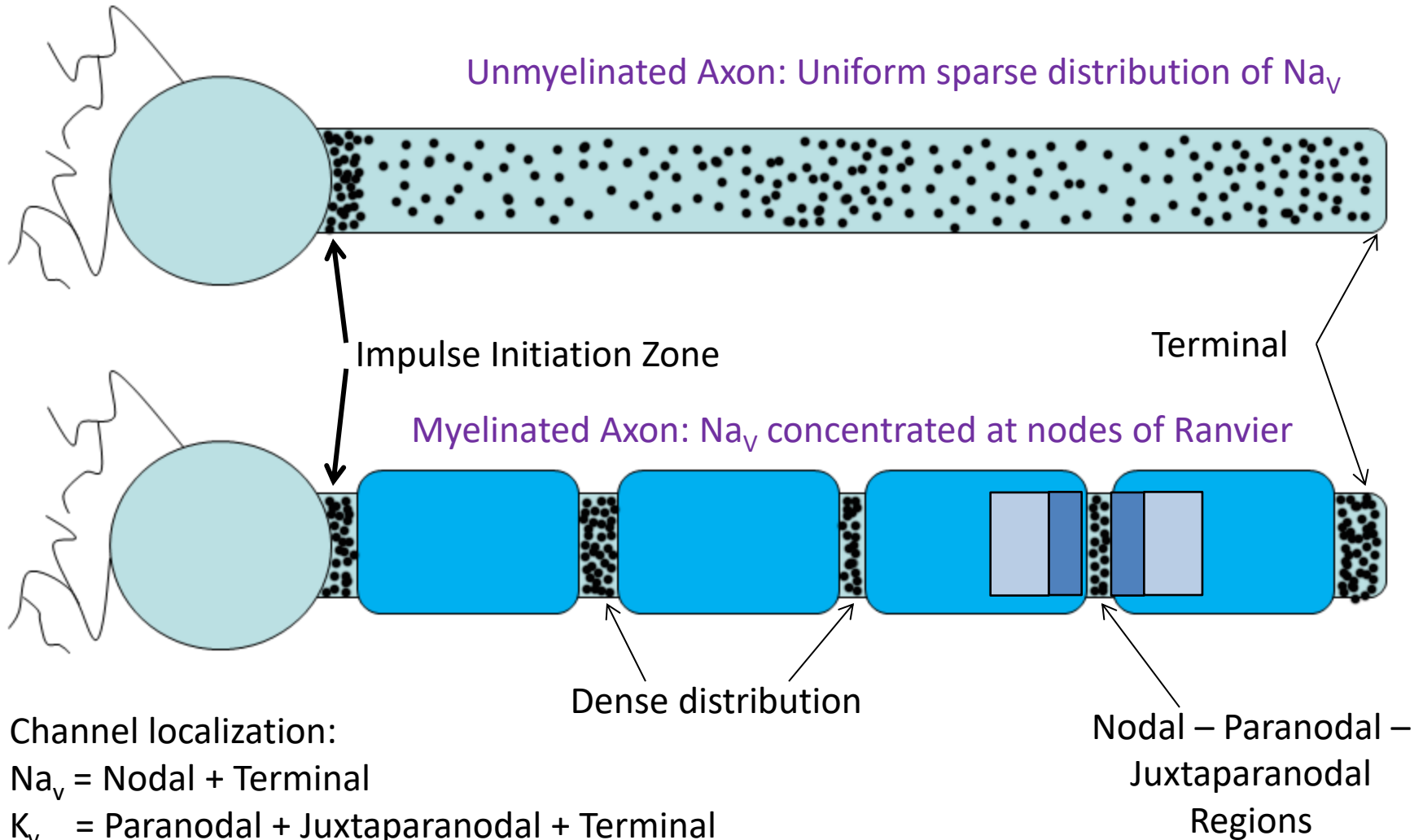
Outward movement of the S4 segment with depolarization also exposes amino acids on the internal S4-S5 links that bind the IFMT (isoleucine, phenylalanine, methionine and threonine) motif in the short D3-D4 intracellular link. After rapid depolarization during action potentials, the bound inactivation (h) gate terminates the influx of sodium.



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Distribution of Voltage-gated Channels



Channel localization:

$\text{Na}_v = \text{Nodal} + \text{Terminal}$

$\text{K}_v = \text{Paranodal} + \text{Juxtaparanodal} + \text{Terminal}$

$\text{Ca}_v = \text{Terminal}$

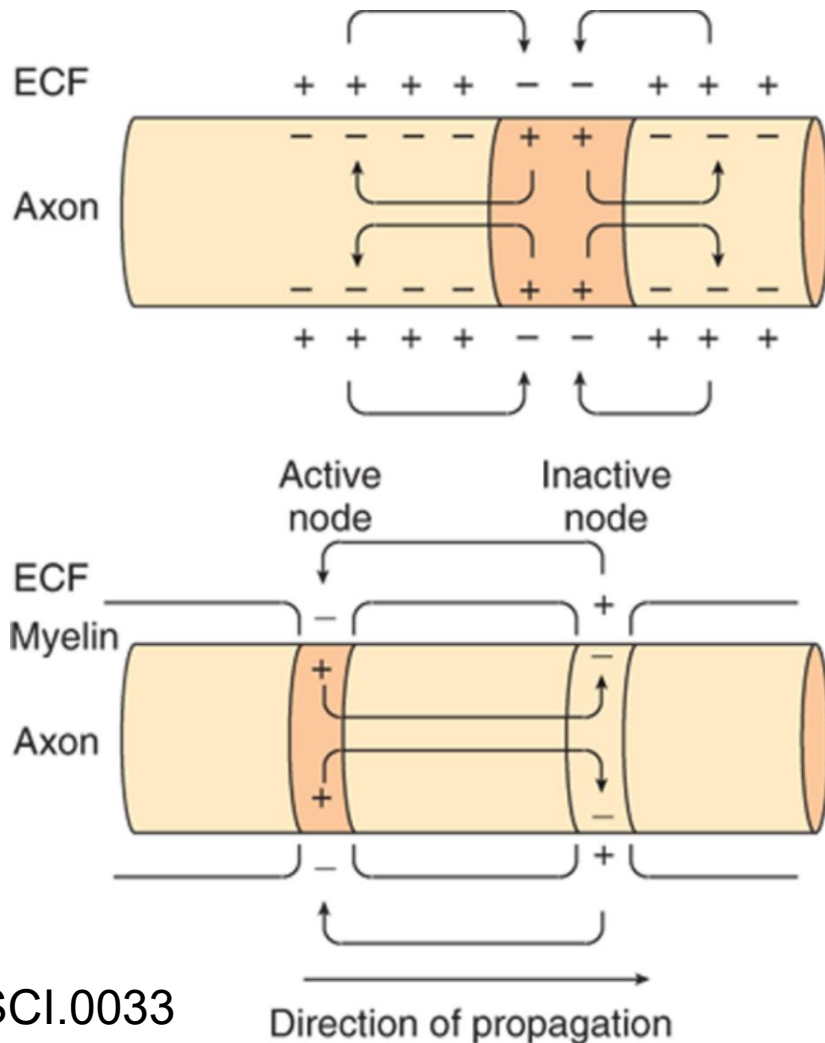
Distribution of Voltage-gated Channels

- Unmyelinated axons
 - Highly concentrated at axon hillock
 - Uniformly but less densely distributed elsewhere
- Myelinated axons
 - Highly concentrated at axon hillocks and nodes of Ranvier

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Rapid Conduction in Myelinated Axons



- Unmyelinated axon (top)
 - Current from depolarized region depolarizes adjacent region, allowing slow propagation of the action potential
- Myelinated axon (bottom)
 - Current from a depolarized node of Ranvier does not readily leak via adjacent membrane, owing to electrical myelin-dependent insulation
 - But the current is sufficient to depolarize the membrane exposed at the next node, allowing the action potential to 'jump' (saltate) from rapidly from node to node

Saltatory conduction

- Myelin sheaths
 - Wrap short axon segments
 - Formed by Schwann cells in PNS, oligodendrocytes in CNS
 - Rich in lipids, act as electrical insulators
- AP 'jumps' from one unmyelinated node of Ranvier to the next, between the myelinated sections
- More energy-efficient and much faster propagation of action potentials

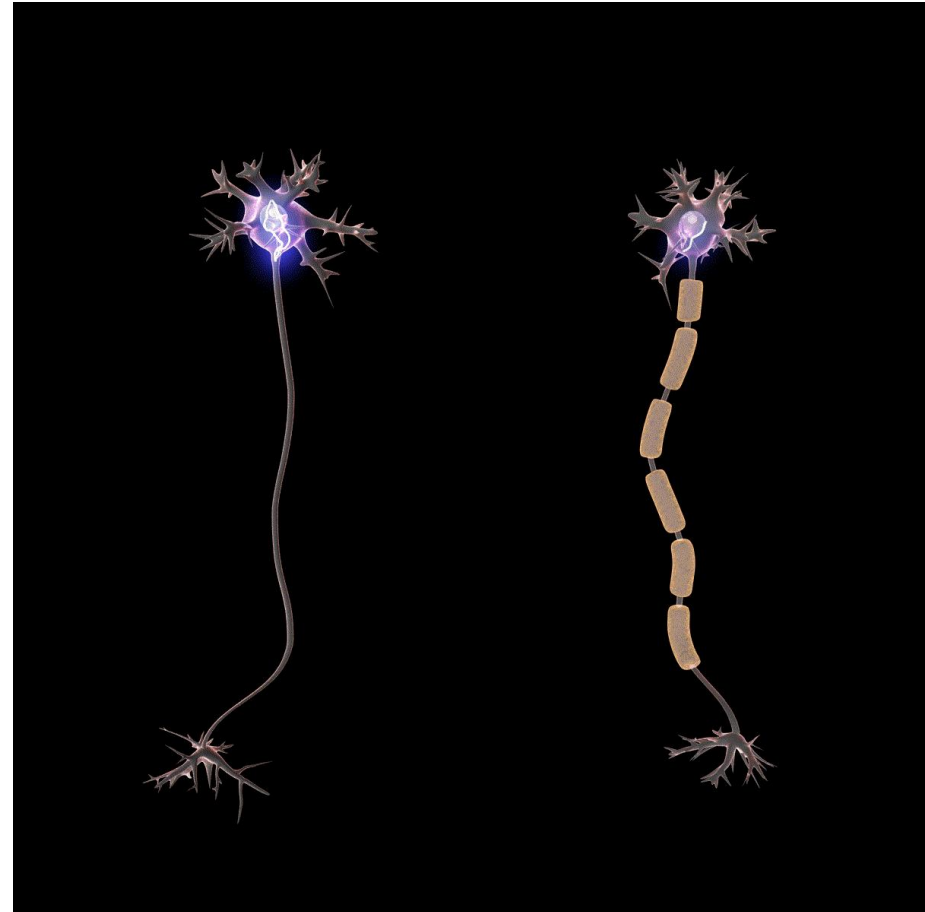


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<https://en.wikipedia.org/w/index.php?curid=41338822>

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Myelination, size and conduction velocity

Type	Erlanger-Gasser	Diameter	Myelin	Conduction velocity	Associated nerve fibres
Ia	A α	13–20 μm	Yes	80–120 m/s	Proprioception, alpha motor neurons
Ib	A α	13–20 μm	Yes	80–120 m/s	Golgi tendon organ
II	A β	6–12 μm	Yes	33–75 m/s	Muscle spindle, cutaneous mechanoreceptors
III	A δ	1–5 μm	Thin	3–30 m/s	Free nerve endings of touch and pressure, nociceptors, cold thermoreceptors
IV	C	0.2–1.5 μm	No	0.5–2.0 m/s	Nociceptors, warmth receptors

Fastest	Large myelinated
Intermediate	Small myelinated
Slowest	Small unmyelinated

Disorders of Myelination

- Diseases that lead to myelin sheath damage, slow or blocking action potential conduction
- Some examples:
 - CNS
 - Myelinoclastic disorders (myelin is being destroyed)
 - Usually autoimmune, e.g. multiple sclerosis (MS)
 - Leukodystrophic disorders (myelin is not properly produced)
 - e.g., CNS neuropathies due to vitamin B12 deficiency
 - PNS
 - Guillain–Barré syndrome or chronic inflammatory demyelinating polyneuropathy, both autoimmune conditions targeting Schwann cells
 - Symptoms include muscle weakness, loss of muscle control, ataxia, loss of sensation, paresthesias (abnormal sensations)
 - Diagnostic methods include tendon reflex testing (e.g. patellar or Achilles tendon), MRI (to detect larger lesions), and electrodiagnostics such as nerve conduction velocity studies

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Effects of local anaesthetics on Na_v

- Lidocaine binds to the open pore with high affinity
- Lidocaine prevents the return of the Domain III voltage sensor to the resting state after repolarization
- Net effect is a preferential Na_v block in actively firing neurons where open Na_v channels are available for binding lidocaine

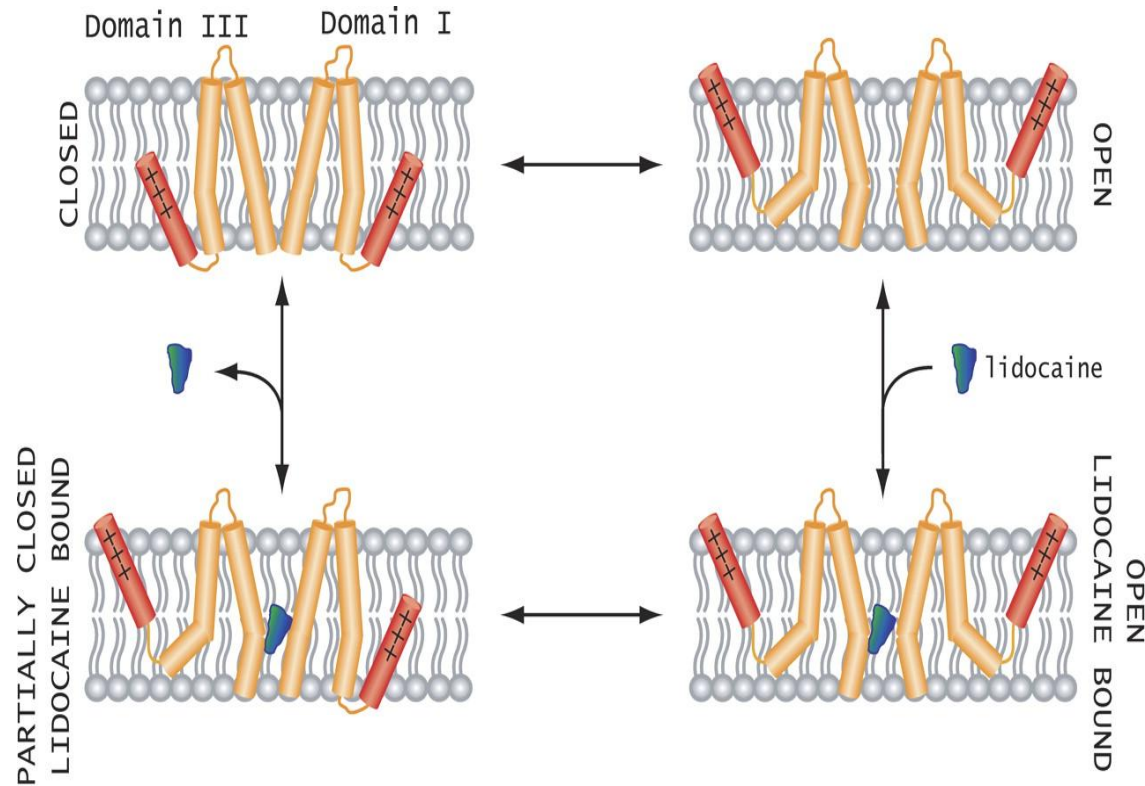
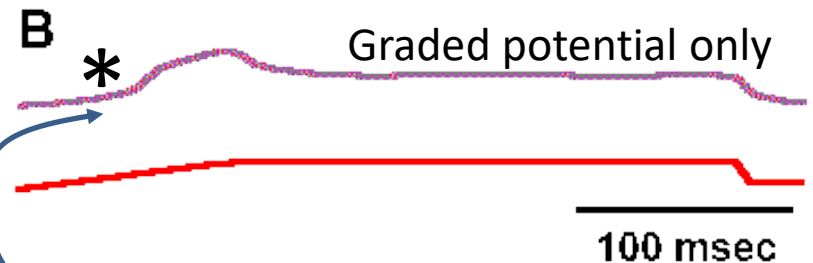
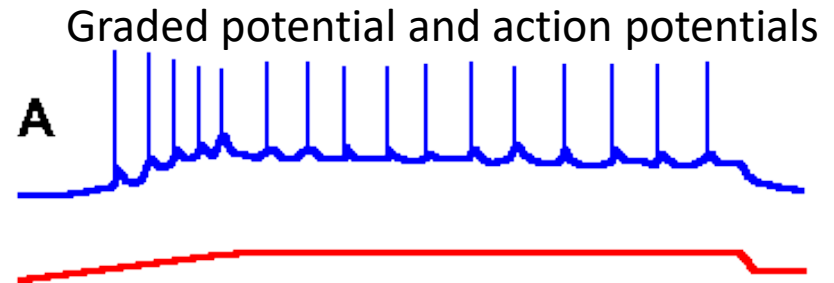


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Suppression of Action Potentials by Lidocaine

- Blue trace (A)
 - Action potentials in afferent during muscular stretch
- Purple trace (B)
 - Absence of action potentials due to lidocaine-induced blockage of voltage-gated Na^+ channels

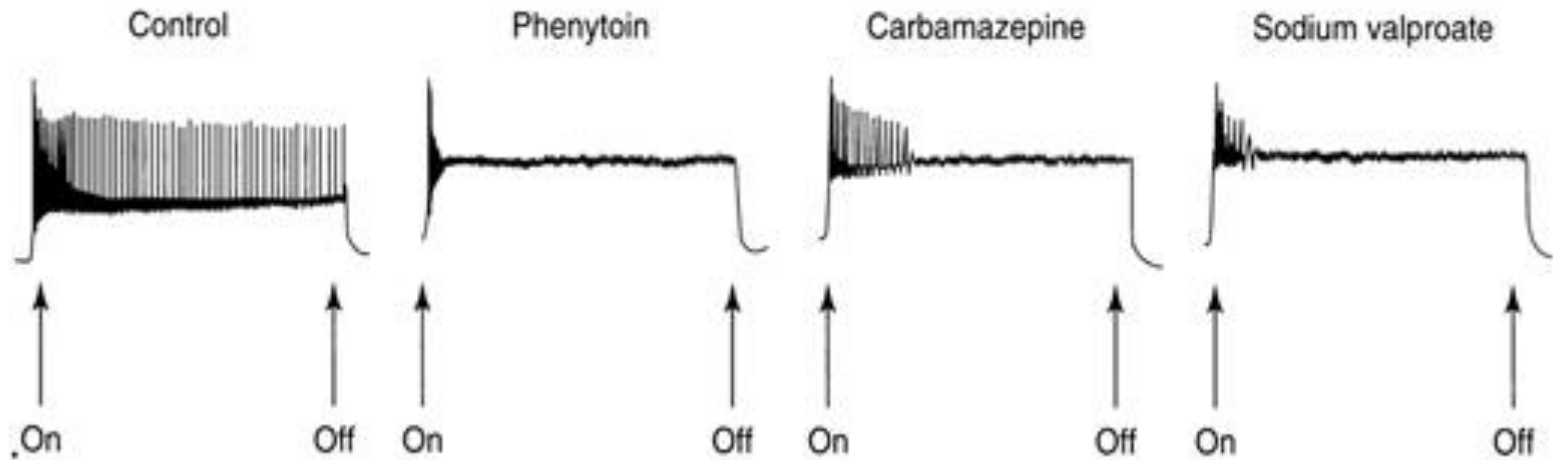


*Local anesthetics (e.g., lidocaine) block action potentials but not graded potentials

Clinical implications

- Local anesthetics may preferentially act on firing and depolarized neurons
 - ‘Use-dependent’ suppression of action potentials
 - Selective for rapidly firing neurons

Related anti-seizure drugs



- Bind to and stabilize inactive configurations of Na_v , blocking Na^+ flux
- Action is 'use-dependent' because they preferentially affect firing or depolarized cells (more inactivated Na_v channels present)
- Similar to local anesthetics like lidocaine
 - Specific binding preferences may differ but common effect is suppression of trains of action potentials
 - Important anti-seizure drugs
 - Dose-dependent blocking of pain but not normal sensation
 - Can stabilize mood